Teacher:	Ray Joshua Balingkit	**Subject:**	Design & Technology
Week:	6 (Term 1)	**Year:**	10–11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will explain how simple mechanical devices transmit or change motion and force. They will develop their understanding through a teacher explanation, a worked example and an evidence-based classroom task. In the second lesson, students will build and test a low-risk model of a simple mechanism. Their recorded reasoning, practical evidence and final checks will show whether they can apply the ideas to a defined user, product or design problem.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher questioning, labelled diagrams, case comparisons and the completed independent tasks. Check accurate terminology, evidence linked to the user or brief and a reasoned explanation of choices. Review recorded tests or traces alongside conclusions, and use the final check in both lessons to identify the next skill needing practice.

LESSON SLOTS - TUESDAY AND WEDNESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	29 September 2026	By the end of this lesson, students will be able to explain how simple mechanical devices transmit or change motion and force.	**H:** Demonstrate a card lever using a ruler and a stable classroom pivot. Ask what changes when the effort is applied farther from the pivot while the load stays in the same place. **ID:** Introduce input and output motion, pivot, effort and load. Use the book to compare levers, gears, pulleys and cams at an introductory level. Explain rotary, linear, reciprocating and oscillating motion, and distinguish changing speed or direction from creating energy. **WD:** Model a lever with labelled effort, load and pivot, then sketch two meshing gears and mark opposite rotations. Explain qualitatively how a smaller driver turning a larger driven gear reduces output speed while increasing available	**Support:** Use colour-coded input and output arrows and manipulable diagrams. Focus first on a lever and a pair of gears before comparing a cam or pulley. **Extension:** Students compare two lever arrangements for the same load or explain the effect of adding an idler gear, distinguishing direction from speed ratio.

			turning effect, with real systems subject to losses. **YD:** Students annotate three mechanism diagrams, identify their input and output and predict a direction or motion change. Pairs demonstrate one prediction using teacher-prepared card components, then individually explain how the arrangement would help a user in a simple product. **C:** Ask students to explain why a mechanism does not provide free energy. Check motion labels and gear directions, then use an incorrect diagram to expose a common misunderstanding.	
2	30 Septemb er 2026	By the end of this lesson, students will be able to build and test a low-risk model of a simple mechanism.	**H:** Show a finished card lever that moves a small sign and ask where it might fail. Invite predictions about the pivot position, card stiffness and unwanted rubbing. **ID:** Explain the modelling brief: create a manually operated card mechanism with an observable input and output. Agree success criteria for movement, stability and repeatability. Use pre-cut card and teacher-prepared holes; students use only the approved fasteners and classroom tools demonstrated. **WD:** Assemble a lever using prepared parts, label the pivot and input and test its movement. Model recording a fault such as rubbing against the base, changing one feature and retesting rather than changing several things without knowing which helped. **YD:** Pairs assemble a	**Support:** Provide pre-cut parts, numbered assembly images and a testing table. Help students locate the pivot while leaving them responsible for predicting the effect of their chosen adjustment. **Extension:** Students compare two pivot positions under the same test conditions and explain the trade-off between input movement, output movement and effort.



			card mechanism, sketch its input and output and test it over repeated operations. They record one fault, make one justified change and repeat the same test. Each student writes how the mechanism works and whether evidence supports the agreed success criteria. **C:** Compare the prediction, first test and revised model. Check that explanations identify the moving parts and the reason for a change, and that students return materials and report damaged components.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel GCSE (9–1) Design and Technology revision notes: 1.5 Mechanical devices
- Teacher computer, display screen and whiteboard markers
- Prepared product scenarios, diagrams and user briefs aligned to the lesson objectives
- Prepared analysis worksheet, comparison grid or test-recording table
- Paper, pencils, rulers, card templates and approved classroom stationery
- Teacher-prepared samples or component cards; pre-cut model parts where required
- Lesson success criteria, observation checklist and labelled evidence folders



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?